

Telstra AWS Agentic Framework (TAAF)

May 2026

Telstra Account team
Amazon Web Services

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What are we solving



Manual efforts to get working in Cloud (rather than automated & repeatable efforts)



Too many tickets & Queue wait times, due to silo'd processes that inhibit moving at speed



The delivery of business outcomes by Workload owners slowed down by work heavy LCM activities

Modernize & Automate using TAAF



Agents built for Telstra's operating env



Agentic AI Tools and Accelerators



CF30 Enabler / Fabric Integration



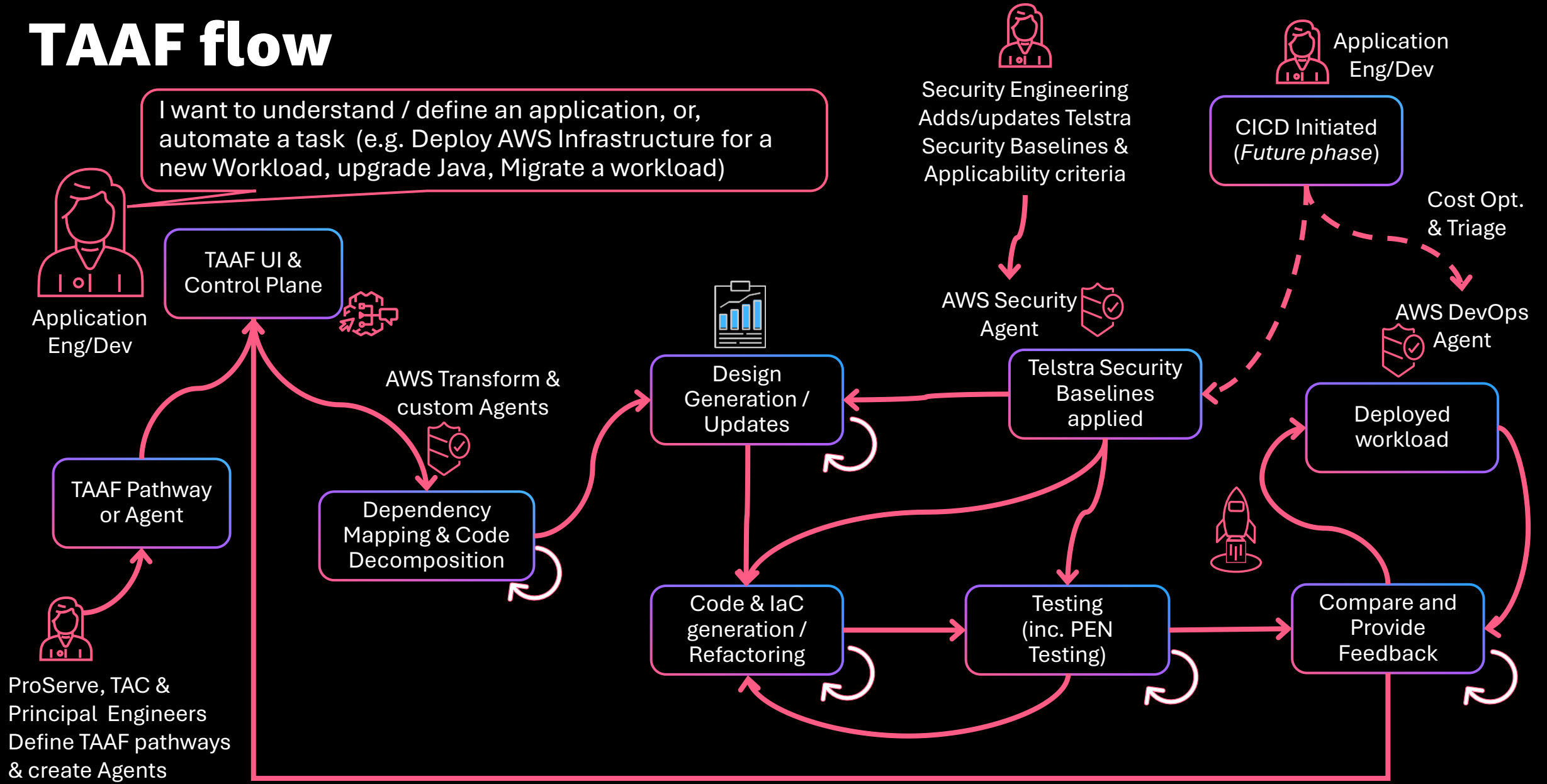
Automate LCM & Optimisation in the Pipeline, not one-off efforts



Agentic Security Reviews & PEN Testing by Engineers/Devs

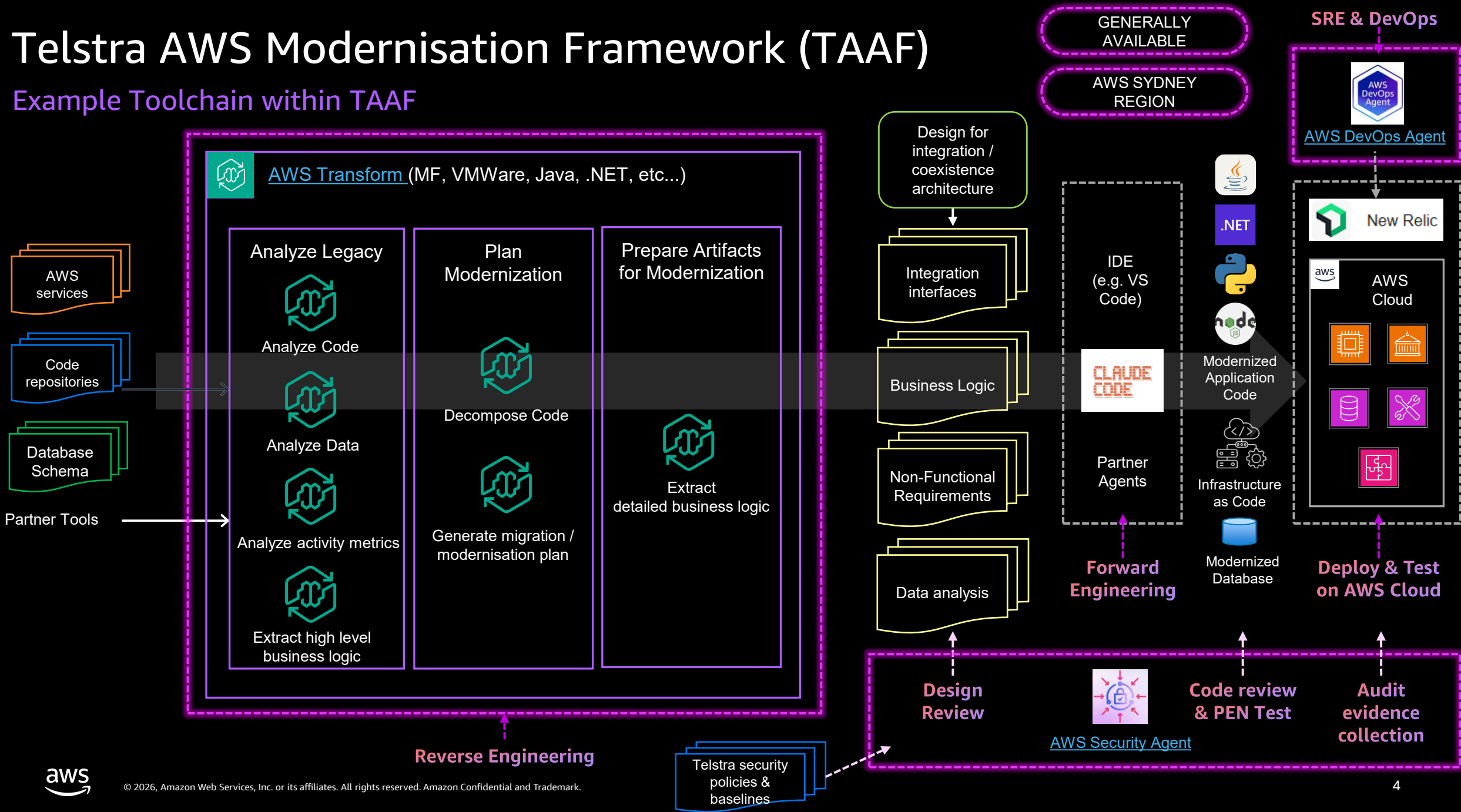
TAAF flow

I want to understand / define an application, or, automate a task (e.g. Deploy AWS Infrastructure for a new Workload, upgrade Java, Migrate a workload)



Telstra AWS Modernisation Framework (TAAF)

Example Toolchain within TAAF



Why do this

BEFORE — Manual & Constrained

Knowledge on workloads has gaps

Teams have inherited workloads from previous vendors/teams and often have only rudimentary knowledge of the code/logic/IP. Documents are scattered, so new Dev's take 6+months to learn their workloads

Solution and Onboarding

Solution Architecture Document (SAD) creation takes weeks; manually reviewed against ~100 Cloud Security Baselines.

Every request is unique and manually Assessed & Approved

This applies for the 21 step Cloud onboarding process, including workload heavy processes for Security/ Identity / Firewalling, FinOps, Accounts/Databases/Container env's and Cyber Assurance

Cyber Assurance

3-6 week wait for resource assignment; driven by meetings, email and document reviews

Security & Compliance

3-5 weeks and \$10k+ per manual PEN test; so most apps tested once
One-off PEN test at ATO/ORC only; 81% knowingly deploy vulnerable code to meet deadlines

AFTER — Agentic & Self-Service

App Discovery and SAD/BRE generation automated

AWS Transform automatically discovers, documents and extracts business logic in hours, using documents, source code or deployed configurations

Detailed and Accurate documentation to base automation on

80% of SAD production and updates to align to Telstra's Security Baselines can be delivered 1-2 days (as shown during Telstra's Afterburner trial of the Security Agent)

DevSecFinOps largely automated to Remove Time in Queues

Low cost or common workloads could be auto approved and provisioned. In future phases, integration to the CICD pipeline will automate Cost & Security checks during ongoing Code Releases

Cyber Assurance

Developers are aligned to baselines automatically during Design & Code reviews; addressing security risks early & producing evidence to speed up Cyber reviews (enabling a shift to "by exception" approach in the future)

Security Testing

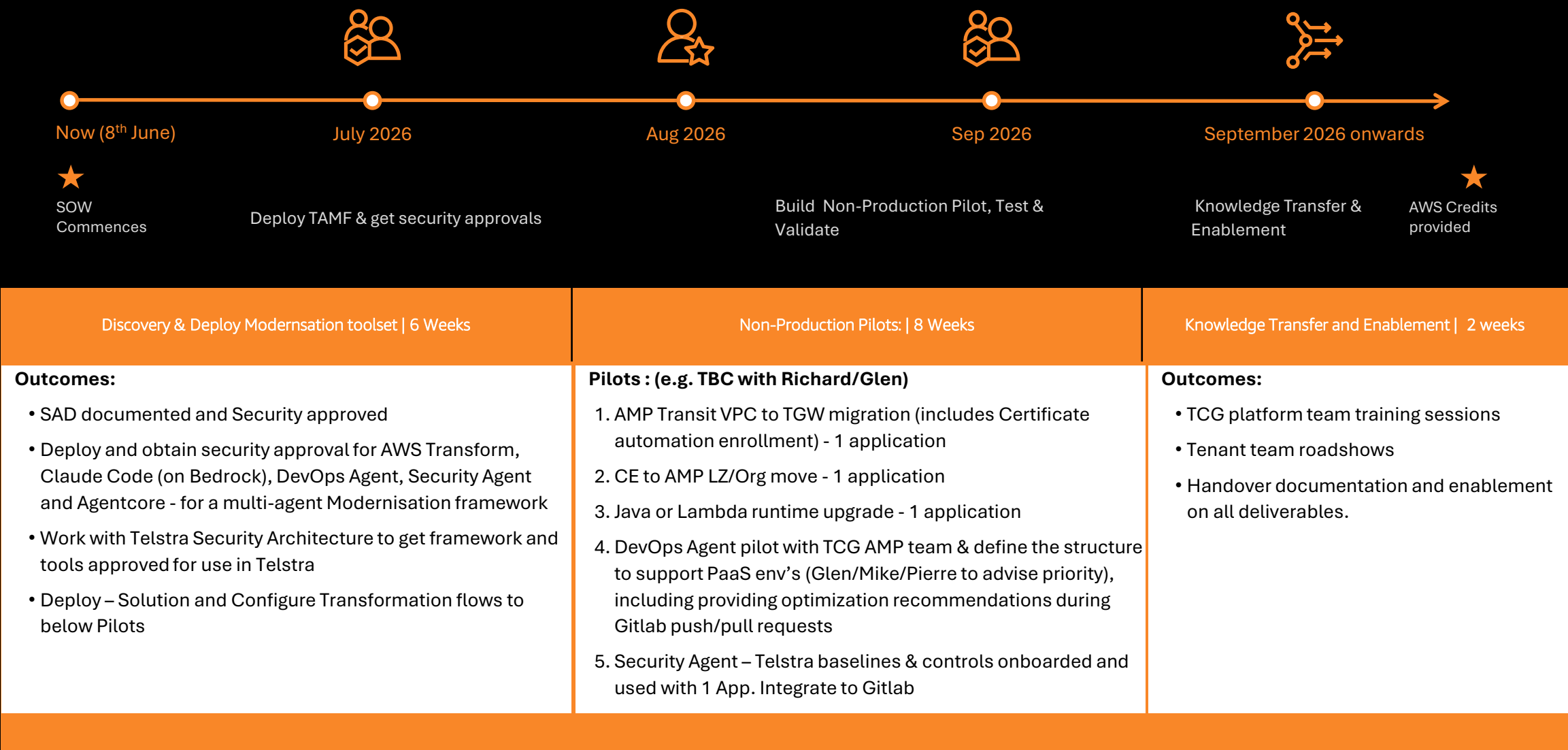
<7 hour AI PEN test with self-adaptive attack chaining and novel vulnerability discovery, enables on-Demand PEN tests at \$300-\$500 per test can be done any release, providing audit evidence with detailed proof



Telstra today vs. Telstra with TAAF — real data from the Afterburner Security Agent trial



Deployment Timeline



Estimated Costs for TAAF & AWS Investments

AWS Credits will offset the TAAF component usage until at least 31 Nov 2026

Estimate (TBC in Pilots): \$250 - \$550 per pathway executed (after credits are exhausted)



AWS ProServe

\$0 (SOW Costs ← PPA Credits)

The \$1.5M PPA Modernisation credit will be used to offset the \$1.51M AWS ProServe SOW costs. Telstra to engage ProServe via Fieldglass and AWS to provide PPA credits once SOW is complete. SOW Invoices to start July/August, so cost & credits will both occur in FY27.



AWS Security Agent

AWS credits offsets to 31 Nov

300 free task-hours/month will be offset by AWS credits, for Penetration tests using the AWS Security Agent until 31st Nov 2026. Provided to Telstra as thank you for feedback during the Agent's preview period. After 300 hours/month, \$50 per task hour applies (\$200-\$350 per PEN test estimated & tests delivered in hours compared to weeks currently).



AWS DevOps Agent

AWS credits offsets ongoing

As Telstra is an AWS Enterprise Support customer, Telstra is already receiving ~\$183k per month of AWS credits which can be redeemed to offset the DevOps Agent costs (these credits can only be used for the DevOps Agent and expire monthly if not used. At \$0.0083 per agent-second, it is unlikely Telstra will consume >25% of these credits. TBC during Pilots).



Amazon Bedrock & Bedrock AgentCore

AWS PPA GenAI credits

The AWS Private Pricing Agreement (PPA) for Telstra includes "GenAI credits", to offset a significant amount of usage via Amazon Bedrock and Amazon Bedrock AgentCore. Ian Terry/Nikolai have details.



AWS Transform

No Cost for default Agents

AWS Transform has no charge for the default agents for functions like .NET, Java, Mainframe, VMWare, etc... transformation. If custom agents are used within Transform, then \$0.035 / agent minute applies. No custom agents are anticipated at this stage.



Claude Code on Bedrock

AWS PPA GenAI credits

Claude Code on Amazon Bedrock meets Telstra Security requirements and results in LLM token usage charges via Amazon Bedrock. So the PPA GenAI credit offsets apply.

COMMERCIAL IN CONFIDENCE (All Figures in USD)



Suggested next Steps

1. Richard/Glen confirm the desired Pilots (by 13th May)
2. AWS to send draft SOW (by 15th May)
3. Amend PCF SOW to use remaining budget and get moving quickly
 - Resource onboarding, access & laptops could start using existing PCF SOW
 - SAD generation (AMP Account setup & Security approvals) to use remaining funds on PCF SOW
4. Get TAAF SOW approved & work continues (by 8th June)

Note – TAAF SOW won't invoice until July/August, so ProServe costs and credits would both be in FY27

Appendix



AWS ProServe Scope & Timeframe



Summary for TAAF Deployment SOW

This proposal outlines the engagement plan for deploying, securing, and operationalising a multi-agent Modernisation framework within Telstra. The solution will leverage services like AWS Transform, Claude Code or Kiro, Amazon Bedrock Agentcore, Enterprise AI Hub, and Accelerator Hub along with Security/DevOps agents. The engagement includes non-production pilots across key Modernisation use cases and concludes with comprehensive knowledge transfer and enablement of Telstra teams.

Key Outcomes:

- Modernisation Platform Operational – AWS Transform, Claude Code or Kiro, Amazon Bedrock Agentcore (multi-tenant), and Enterprise AI Hub deployed, security-approved, and production-ready with App/Cloud discovery and New Relic NLQ observability integration
- Non-Production Pilots Validated – Four pilots executed and validated — AMP Transit VPC→TGW migration (1 app), CE→AMP LZ/Org move (2–4 apps), Java/Lambda runtime upgrade (1 app), and DevOps/Security agent pilot — with documented outcomes and test results
- Gen AI Accelerated to Production – Accelerator Hub deployed with 1–2 key Gen AI use cases identified, built, and promoted to production, demonstrating tangible business value
- Multi-Agent Framework Proven – Security and DevOps agents configured, piloted, and validated within the multi-agent Modernisation framework, enabling scalable agent-driven operations
- Self-Sufficiency Achieved – Platform team trained, tenant teams enabled via roadshows, and complete handover documentation delivered for independent operation and extension of all capabilities



Telstra AWS Modernisation Framework (TAAF)

Opportunity

Empower Modernisation at scale for Telstra, using a multi-agent and self-service capability to drive efficiency, enhance security and governance on existing and future applications.

The Big Idea

- Shift from point-in-time modernisation efforts to framework-based approaches
- Deploy extensible & reusable toolsets, to solve multiple problems
- Focus on repeatable patterns rather than one-off activities
- Use modernisation SoW as vehicle to implement solution capabilities
- Deliver shared team workspaces within TAAF for collaboration and reuse of generative AI use cases

Scope

- Deploy and Obtain security approval for AWS Transform, Claude Code or Kiro and Agent-core (multi-tenant flavour) and Security/DevOps agents (configured with multi-agent modernisation framework, similar to the [CBA implementation shared at ReInvent 2025](#))
- DevOps integration with New Relic (Chatbot style NLQ's into observability)
- Security agent pilot
- Co-develop validation criteria with Telstra
- Deployment of Generative AI workspaces for team collaboration
- Deploy Bedrock AgentCore
- Program Governance

Non-Production Pilots

- AMP Transit VPC to TGW migration (Includes Certificate automation enrolment) - 1 application
- CE to AMP LZ/Org move - 1 application
- Java or Lambda runtime upgrade - 1 application
- - DevOps and security agent Pilot
- Knowledge Transfer & Enablement
- TCG platform team training sessions
- Tenant team roadshows
- Handover documentation and enablement on all deliverables

RACI proposed

Squads made up of:

- AWS Modernisation Tech Lead, Cloud Architects, DevOps Specialists, Gen AI Engineers/ Architects, Governance and Security
- Modernisation support vendor (e.g. Versent) for Modernisation and AI/ML uses cases
- Telstra Infra SME for assessment tool deploy
- Telstra Gen AI / AIML Team

- AWS Team

- Cloud/Network Architects
- Application Modernisation Specialists
- DB Modernisation Specialists
- Security Specialist
- AI/ML Engineers

Telstra Team (Part-time)

- AMP & CE Engineers
- Application Engineers for Pilot Apps
- Security Arch
- Gen AI Engineers

Key Dependencies

- AWS account access and required IAM permissions will be provisioned by TCG within Week 1–2
- Security approval process timelines are best-effort; delays in Telstra security review may impact Phase 3 completion (note – existing Bedrock & Claudecode on Bedrock approvals will help fast-track this)
- Pilot applications have been pre-identified or will be confirmed no later than end of Week 2
- New Relic environment is available with appropriate API access for DevOps agent integration
- Non-production environments are available and representative of production configurations
- TCG will provide dedicated points of contact for each pilot workstream

Please refer to the next slide for more detail



Telstra AWS Modernisation Framework (TAAF)

Kick off, Discovery & Planning (Weeks 1-2)	Assessment & Design (Weeks 3-6)	Deployment & Validation (Weeks 7-14)	KT & Handovers (Weeks 15-16)												
Focus: Alignment, access, baselining Activities - ProServe will kick off project & deep dive on enablers & dependencies - Establish project governance, communication cadence, and RACI - Gain access to environments, tooling, and relevant documentation - Baseline current state of target applications, infrastructure, and security posture Outcomes - Detailed delivery plan, including dependencies, milestones and showcases. - Discovery Report – Current State (Applications, Cloud, AI/ML Landscape, Dependencies) - Identify any Telstra process or governance approvals that may impact delivery - Identify high level success criteria - Environment Access & Readiness Checklist. - Gen AI Use Case Shortlist (1–2 selected for Accelerator Hub) <table><tr><th>Telstra Role</th><th>Effort</th></tr><tr><td>Telstra Application Owner</td><td>25%</td></tr><tr><td>Telstra Security Architect</td><td>50%</td></tr></table>	Telstra Role	Effort	Telstra Application Owner	25%	Telstra Security Architect	50%	Focus: Architecture, pilot planning, security review Activities - Complete detailed technical assessment of pilot applications and target environments - Design deployment architecture for TAAF (including AWS Transform, Claude code or Kiro, and Bedrock Agentcore (multi-tenant)) - Define integration patterns, agent configurations, and security approval pathway - Develop detailed pilot runbooks and test case frameworks Outcomes - TAAF Deployment Architecture Document (AWS Transform, Kiro, Agentcore) - Multi-Agent Framework Configuration Design, New Relic NLQ Integration – Technical Design Document, AMP Transit VPC to TGW Migration Runbook (incl. Certificate Automation) - CE to AMP LZ/Org Move – Migration Plan (2–4 apps), Java/Lambda Runtime Upgrade – Assessment & Upgrade Plan - Amazon Bedrock Agentcore – Multi-Tenant Architecture & Configuration Design, Enterprise AI Hub – Platform Architecture & Governance Framework, Accelerator Hub – Gen AI Use Case Architecture & Production Deployment Design Master Test Plan & Test Cases (all pilots), Design Review Sign-Off	Focus: Build, deploy, pilot execution, test cases Activities - Deploy and obtain security approval for core platform components - Execute all non-production pilots with structured validation - Conduct comprehensive testing against defined test cases - Iterate based on findings and stabilise deployments Outcomes - Deployed & Security-Approved: AWS Transform, Kiro, Agentcore (Multi-Tenant), Configured & Operational: Security Agent, DevOps Agent (Multi-Agent Framework) - New Relic NLQ Chatbot Integration – Operational - Pilot Completion Report – AMP Transit VPC to TGW (1 app, incl. cert automation), Pilot Completion Report – CE to AMP LZ/Org Move (2–4 apps) - Pilot Completion Report – Java/Lambda Runtime Upgrade (1 app), Pilot Completion Report – DevOps & Security Agent - Gen AI Production Deployment Report (performance, accuracy, SLAs) - Accelerator Hub – 1–2 Gen AI Use Cases Deployed to Production - Test Execution Report (all test cases with pass/fail/observations)	Focus: Training, roadshows, documentation Activities - Enable TCG platform and tenant teams to independently operate and extend all deployed capabilities - Deliver structured training and comprehensive handover documentation - Conduct tenant team roadshows to drive adoption and awareness Outcomes - Training Session Recordings & Materials (Platform Team) - Tenant Team Roadshow Deck & Recordings - Handover Documentation Pack (Operations Runbooks, Configuration Guides, ADRs, FAQs) - Enablement Walkthrough Completion Sign-Off - BAU Support & Escalation Model - Project Closure Report & Lessons Learned <table><tr><th>Telstra Role</th><th>Effort</th></tr><tr><td>Telstra Application Owner</td><td>25% (review, approval)</td></tr><tr><td>Telstra Security Architect</td><td>25% (review, approval)</td></tr></table>	Telstra Role	Effort	Telstra Application Owner	25% (review, approval)	Telstra Security Architect	25% (review, approval)
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aws

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TAAF Agentic Framework - Expected Benefits

Empowering Telstra teams to modernise, secure and operate workloads on AWS — faster and with fewer dependencies



Accelerated Discovery & Current State Analysis

Weeks → Days

Automatically discover and document workloads — code, dependencies and infrastructure. Rapidly develop SADs /As-Built docs from code, IaC or diagrams and show relevant Telstra Security baselines.



Empowered Developers

Eliminate Queue Dependencies

Teams modernise, refactor and secure code using self-service capabilities, reducing delays for prioritization alignment & resource availability from other teams.



Proactive Security

4-7 Hrs vs 3-5 Weeks

AI-powered PEN testing and design review replaces weeks of manual effort. \$10k+ per PEN test, reduces to \$300-\$500 per PEN test.



Continuous Compliance

On Every Code Release

Automated PEN testing and audit evidence on each major release or pull request — not just one-off at ATO/ORC



Reusable Patterns

Build Once, Reuse Many

Standardised toolchain for Java, .NET, mainframe and VMWare modernisation at scale. Shared workspaces for team artefacts. Standards & BP's enforced



End-to-End Automation

Full Lifecycle Coverage

CI/CD pipelines, SRE automation and observability from day one with AWS DevOps Agent

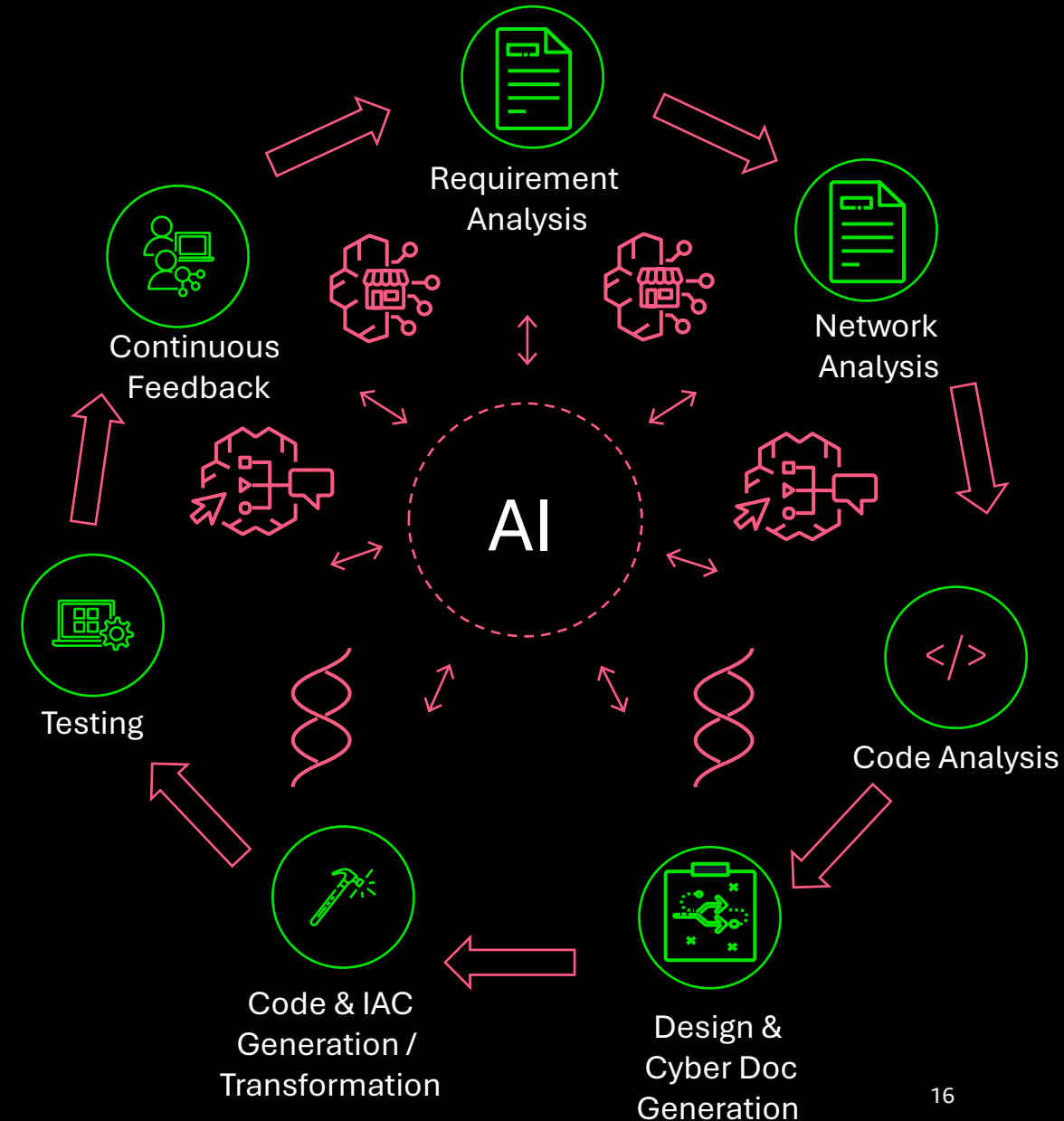
Refer appendix for more details

Appendix



A New Way of Working

- Multiple workflows / pathways powered by AI
- Human-in-the-loop feedback and approvals
- modernization and migration engine powered by custom knowledgebase
- Institutional knowledge capture
- Build high quality detail to drive downstream quality outputs
- Short and long term memory

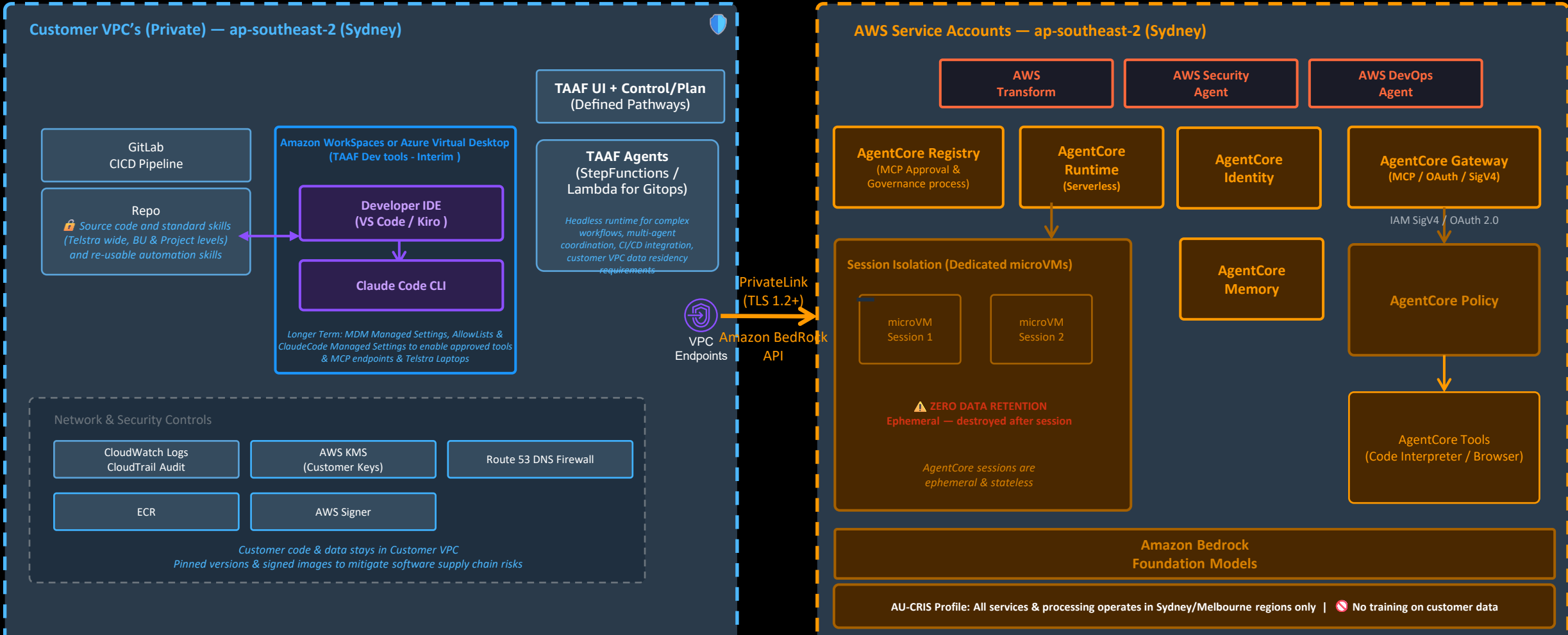


Secure Deployment Architecture Overview

Data Protection & IP Security Controls | AWS Sydney Region (ap-southeast-2) | AU-CRIS Compliant | More detailed doc's available under NDA

AWS Boundary

Customer VPC



- Security & Data Protections

Zero Data Retention (ephemeral microVMs destroyed post-session)

Session Isolation (dedicated microVM per user, cross-tenant isolation)

No Model Training (customer data never used for model improvement)

Customer-Owned Logs (all data plane logs in customer account)

VPC Mode (complete network isolation + DNS Firewall)

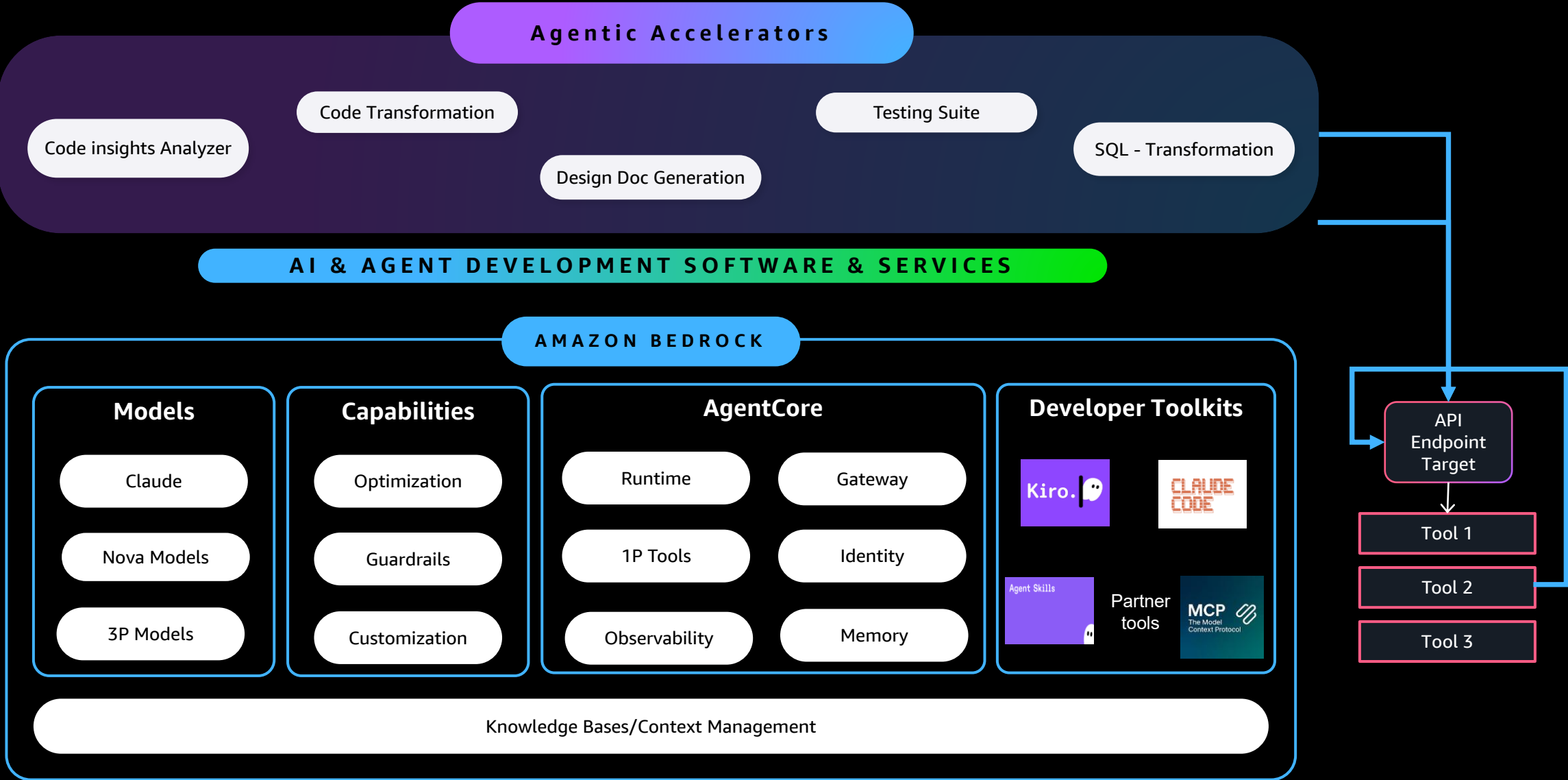
Encryption at Rest (KMS) & In Transit (TLS 1.2+)

Cedar Policy Enforcement (deterministic auth outside LLM reasoning)

AU-CRIS Data Sovereignty (all processing in AU regions only)

No AWS or 3rd Party access

Telstra AWS Modernisation Framework (TAAF)



Agentic AI powered Application Modernization



ReImagine (TAAF)

Before

application stack



Application layer



Database layer

Deployment

**.NET Framework 3.5+
Java 1.8**

UI Layer (Web Forms, MVC Razor)
Business Logic Layer
Data Access Layer

SQL Server, Oracle Server

Schema and stored procedures
conversion
Data migration

**Virtual Machine running on
Windows Server/Linux
Server**

Portfolio of Applications

LARGE-SCALE MODERNIZATION



Code Analysis, Code Transformation

DEVELOPER / DBA LED MODERNIZATION



AWS DMS/Visual Studio IDE extension



AWS Transform-VmWare



AWS Transform-Bedrock –
Wave planning

After

Open-source, cloud-native stack

**Cross-platform .NET 8 & 10
Java 18/22**

UI Layer (Web Forms→Blazor, MVC Razor)
Business Logic Layer (modernized)
Data Access Layer (modernized)

**Amazon Aurora PostgreSQL
RDS**

Converted schema and stored
procedures
Migrated data

Amazon ECS or EC2 Linux

**Application Discovery –
wave planning**



Benefit Details



Accelerated Discovery & Documentation

AWS Transform (MF, VMWare, Java, .NET)

- Reverse-engineer undocumented legacy applications
- Auto-extract business logic, data schemas & dependencies
- Generate migration and modernisation plans
- Replace weeks of manual codebase analysis with hours



Empowered Development Teams

Claude Code on Amazon Bedrock + Kiro

- Developers modernise and refactor code with AI assistance
- Decompose monoliths into microservices
- Generate integration interfaces & Infrastructure as Code
- Remove dependency on specialist queues and bottlenecks



Proactive AI-Powered Security

AWS Security Agent (Validated: Telstra Afterburner Trial)

- Design review: SADs produced in 2 days instead of weeks
- PEN testing: 3.5-7 hours vs 3 weeks at \$10k+ per engagement
- Discovered novel multi-WAF attack vector outside Telstra baselines
- Self-adaptive attack chaining with transparent evidence & reasoning



Continuous Compliance & Evidence

AWS Security Agent + DevOps Agent

- PEN test triggered on every code release, not just ATO/ORC
- Audit-ready evidence collected automatically in hours
- Significantly reduces Cyber Assurance review effort
- Shift-left security: developers align to baselines without being security experts



Reusable Modernisation Patterns

TAAF Toolchain (Reusable Across Teams)

- Java upgrades, .NET migrations, mainframe modernisation
- Standardised reverse and forward engineering workflows
- Embed Telstra security baselines consistently across teams
- Increasing ROI with every subsequent workload



End-to-End Lifecycle Automation

AWS DevOps Agent + AWS Transform

- SRE and DevOps automation embedded from day one
- Automated CI/CD pipeline generation and management
- Observability and monitoring with New Relic integration
- Deploy and test on AWS Cloud with minimal manual handoffs

Transformation Impact

BEFORE — Manual & Constrained

App Discovery

Weeks of manual analysis to document undocumented workloads and dependencies

Design Review

SAD creation takes weeks; manually reviewed against 20+ Cloud Security Baselines

Security Testing

3-5 weeks and \$10k+ per manual PEN test; so most apps tested once

Cyber Assurance

4-6 week wait for resource assignment; driven by meetings, email and document reviews

Compliance

One-off PEN test at ATO/ORC only; 81% knowingly deploy vulnerable code to meet deadlines

AFTER — Agentic & Self-Service

App Discovery

AWS Transform automatically discovers, documents and extracts business logic in hours

Design Review

SAD produced in 2 days; Security Agent iteratively validates against Telstra baselines

Security Testing

<7 hour AI PEN test with self-adaptive attack chaining and novel vulnerability discovery. On-Demand PEN tests at \$300-\$500 per test can be done any release.

Cyber Assurance

Developers are aligned to baselines automatically during Design & Code reviews; addressing security risks early & producing evidence to speed up Cyber reviews (enabling a shift to “by exception” approaches in the future)

Compliance

Continuous PEN testing on every code release; audit evidence with detailed reasoning and proof



Telstra today vs. Telstra with TAAF — real data from the Afterburner Security Agent trial



AWS Security Agent — Telstra Afterburner Trial Results

2-week trial by the Telstra Afterburner team (Happy & Tegan) | March 2026 | 20 Cloud Security Baselines tested

2 Days

SAD Generation

Previously took weeks

3.5-7 Hrs

PEN Test Completion

vs 3 weeks / \$25-50K manual

New Finding

Novel Attack Vector

Multi-WAF / LB threat vector discovered



Agentic Design Review

- AI co-pilot generated SAD from source code in 2 days
- Security Agent identified gaps against Telstra Cloud Security Baselines
- Iterative feedback loop: agent flagged gaps, developer remediated, agent re-validated
- Result: SAD aligned to as-built config and Telstra security standards



Agentic PEN Testing

- 3.5-7 hours for comprehensive security assessment
- Self-adaptive: dynamically selects attack vectors from code and deployment
- LLM-as-Judge validates finding integrity independently
- Transparent logging with detailed reasoning and evidence for each finding



Key Outcomes

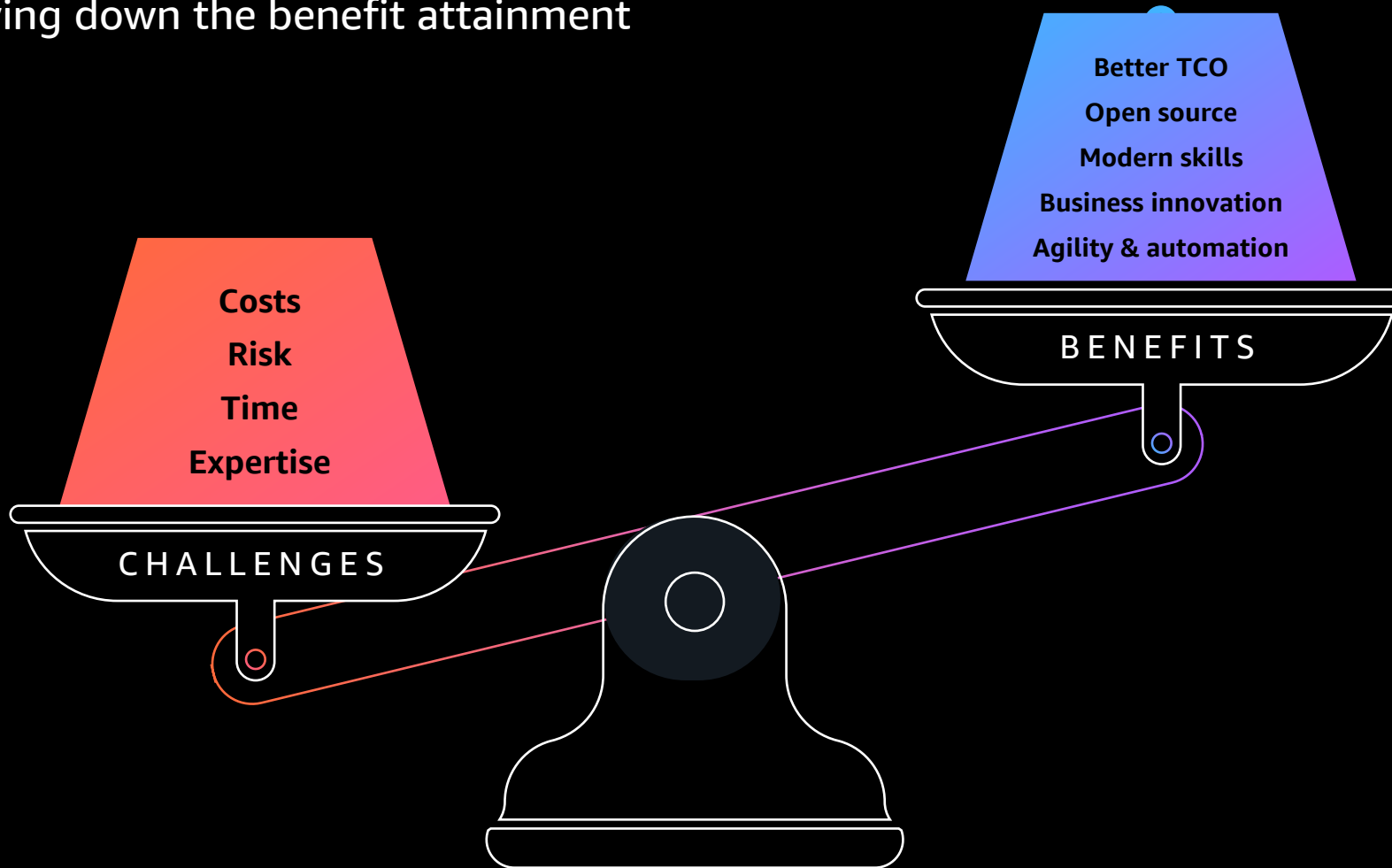
- ✓ Developer Self-Sufficiency: aligned designs and code to Telstra standards without being security experts
- ✓ No Waiting: eliminated dependency on resource-constrained Cyber Assurance teams
- ✓ On-Demand & Audit Ready: continuous PEN testing on each release, not just at ATO/ORC
- ⚠ Surprise: discovered existing baselines does not cover multi-WAF / LB situations (external + LB WAF). This finding was fed back to Security Engineering to determine if a new Baseline is needed for this scenario.

Source: AWS Security Agent trial at Telstra | Afterburner team | March 2026 | Tested against 20 Telstra Cloud Security Baselines



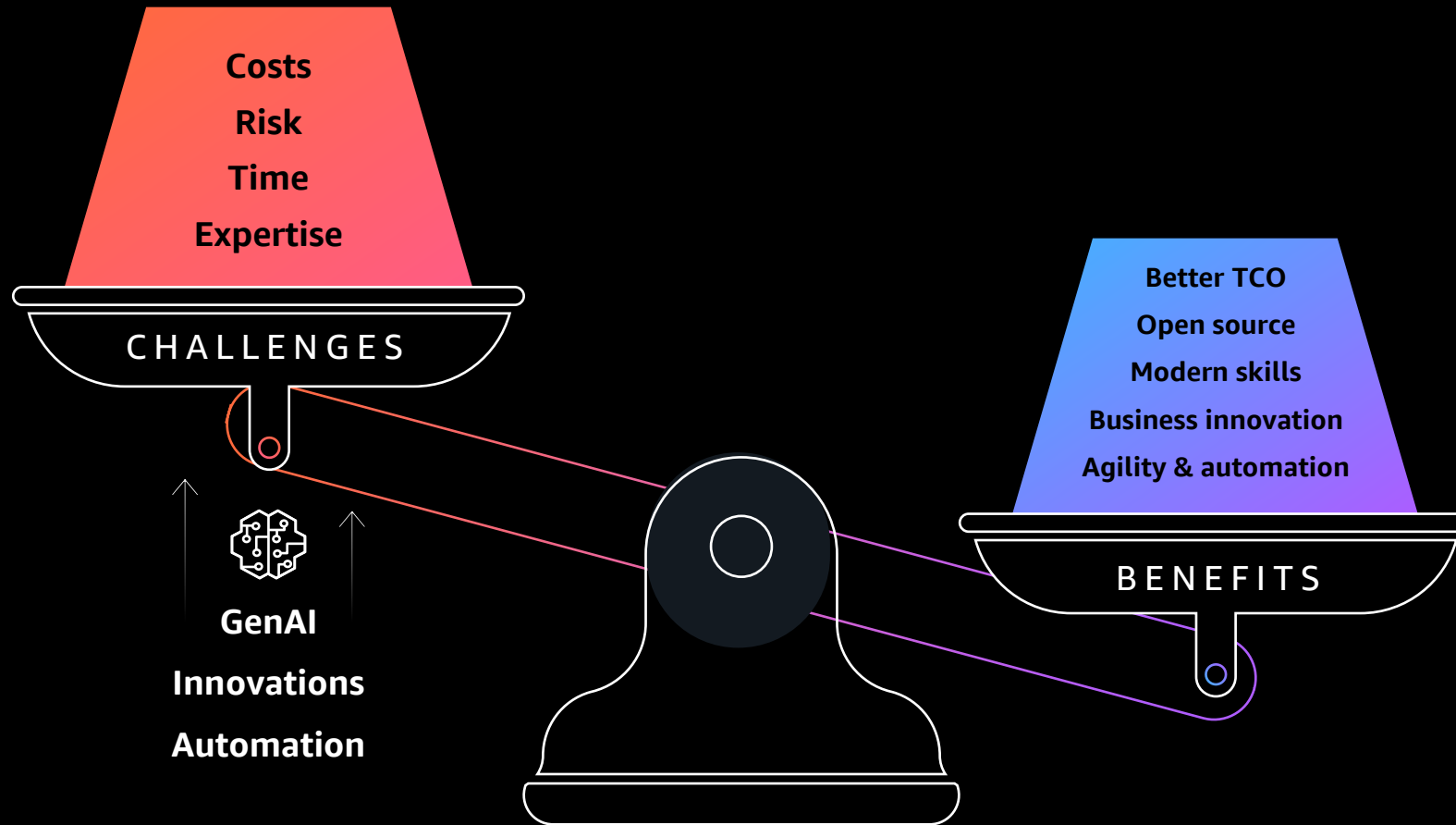
Modernisation: challenges vs. benefits

Challenges slowing down the benefit attainment

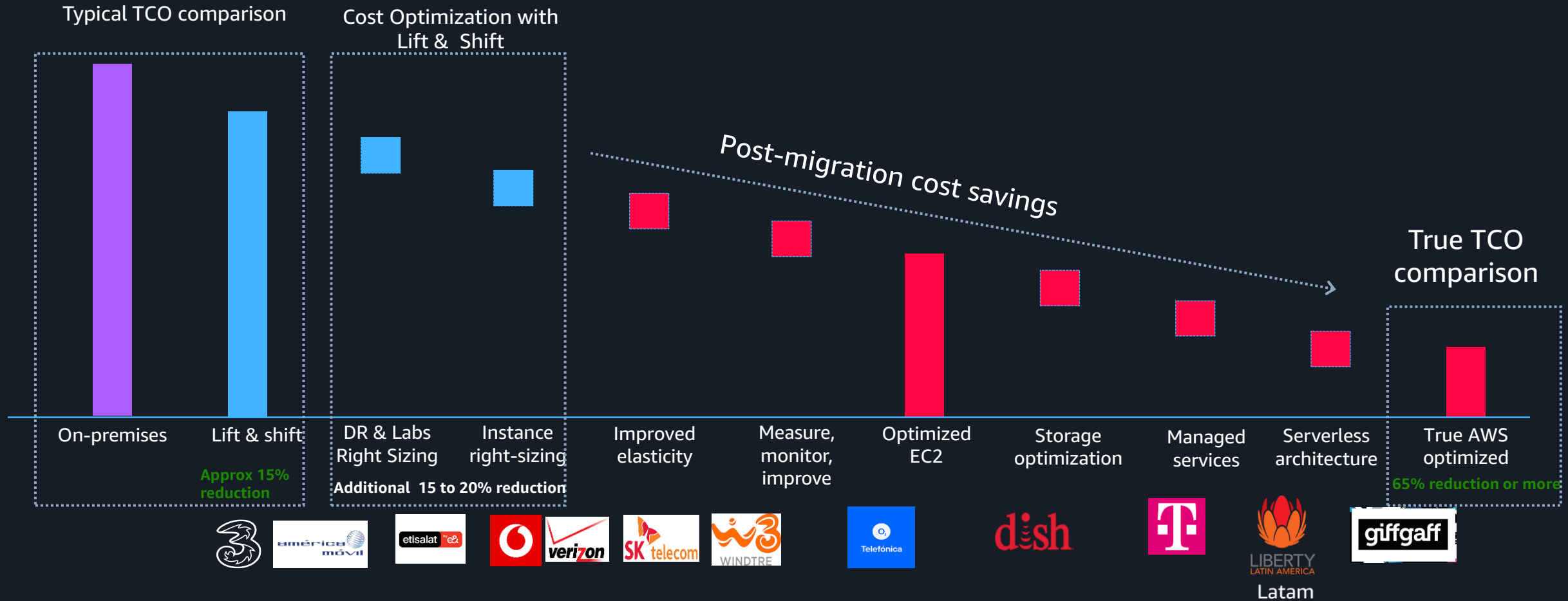


Why now? What changed ?

the advent of Gen AI technology is changing the paradigm



Modernisations on AWS & ongoing Optimisation



Cost Optimization & removal of 3rd party licenses is a Journey

Thank you!

